

KARTIKEYA SINGH

AI / ML Engineer · LLM Systems & RAG · MLOps

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SUMMARY

CS (Data Science) undergraduate who ships ML systems to real users: a production lead-intelligence pipeline for a healthcare startup, evaluated RAG pipelines, and deployed inference services. I test models for failure modes before they ship.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

ML / LLM: PyTorch, Hugging Face Transformers, Scikit-learn, LangChain, ChromaDB, sentence-transformers, BGE reranking, RAGAS, Anthropic API

Serving & MLOps: FastAPI, Docker, Git, MLflow, DVC, pytest, GitHub Actions, REST APIs

Data: Pandas, NumPy, Apache Kafka, Delta Lake, Google Maps / Sheets APIs

WORK EXPERIENCE

AI Engineer (Contract) · Orvica Labs — healthcare SaaS startup

2026 – Present

Python · FastAPI · SQLite · Anthropic API · Google Maps & Sheets APIs · Apify · Pytest

- Built and deployed a lead-intelligence pipeline that autonomously generates **150+ qualified leads every Monday at 6 AM IST** — clinic discovery, enrichment, scoring, deduplication, and CRM sync — contributing to **5–10 paying-client acquisitions within 2 weeks** of launch.
- Engineered multi-stage enrichment combining Claude-based metadata extraction with live booking-platform data, degrading gracefully on upstream API failures.
- Built rule-based lead scoring from operational signals (booking availability, reviews, specialty, website quality) to rank highest-intent prospects first.
- Hardened for unattended production runs: failure isolation, cost caps, structured logging, schema-safe migrations, and **460+ automated tests** in CI.
- Currently building a WhatsApp-based dental intake assistant (DPDP-compliant, emergency-interrupt logic, dentist-facing summary cards) for the same client's clinic platform.

Data Science Intern · Jyesta [\[github\]](https://github.com)

Aug 2024 – Oct 2024

PyTorch · Hugging Face · MLflow · DVC · FastAPI · Docker

- Fine-tuned BERT-base on a balanced 50K Amazon Reviews set for 3-class sentiment, reaching **0.80 macro-F1** — up to 6.4 points over Logistic Regression and BiLSTM baselines, trained with mixed-precision AMP.
- Implemented MLflow experiment tracking and DVC data/model versioning for a fully reproducible train-evaluate-release workflow.
- Shipped a Dockerized FastAPI inference service (single + batch endpoints, health checks, Swagger docs) with a CI-run API test suite.

Data Science Intern · Corteva Agriscience [\[github\]](https://github.com)

May 2024 – Jul 2024

PyTorch · OpenCV · FastAPI · Docker

- Built a multimodal crop-disease classifier (leaf images + soil & weather data): **87% macro-F1 across 12 classes**, a 15-point gain over the 72% image-only baseline for early-stage detection.
- Contributed to soil-health and pesticide-advisory modules for a 50-farmer field pilot in Uttar Pradesh; agronomist feedback drove two retraining cycles and a 4-point gain in minority-class recall.
- Served a disease-risk scoring service over live soil/weather APIs from 3 states via FastAPI in Docker at **<200 ms median latency**.

PROJECTS

EarningsIQ — SEC Filing Financial Q&A (RAG) · [\[github\]](https://github.com)

Python · Claude API · BGE Embeddings · ChromaDB · Cross-Encoder Reranking · Pytest

- Built an end-to-end RAG system over SEC 10-K filings: ingestion, chunking, BGE embeddings indexed in ChromaDB, two-stage retrieval, grounded generation, and evaluation.
- Added cross-encoder reranking over top-20 vector retrieval, lifting **Precision@5 from 0.57 to 0.70 (+23% relative)** on a curated financial-QA benchmark (Recall@20 = 1.00, MRR = 0.82).
- Implemented citation validation, anti-hallucination safeguards, and abstention handling; CI quality gates with Ruff, MyPy, and a full pytest suite.

Streaming Data Warehouse (Kafka → Delta Lake) · [\[github\]](https://github.com)

Python · Apache Kafka · Delta Lake · Docker

- Built a containerized Kafka → Delta Lake streaming pipeline standardizing ingestion across heterogeneous sources, with batch + incremental processing, automated retries, and structured logging.

EDUCATION

B.Tech, Computer Science Engineering (Data Science) · Bennett University · CGPA 8.34/10

2023 – 2027